CONCEPT DOCUMENT

SmartKeep

A priority-based housekeeping task manager that sends the most urgent hotel job to the nearest free housekeeper.

> **Problem statement.** Hotel housekeeping staff may not have a clear priority list for room cleaning. Build a housekeeping task-management tool that prioritizes rooms based on checkout, new check-in, and guest requests.

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Based on	Market Requirements Document and Product Requirements Document v1.0
Stage	[Working prototype / prototype — confirm before submitting]
Date	31 October 2026

1 The problem

In mid-size hotels, housekeeping work is assigned by paper room sheets, phone calls and WhatsApp messages. Nobody has a live list of which job matters most right now, so urgent and time-bound work waits behind routine work.

- **Arrivals wait.** A checked-out room needed for a guest arriving in an hour sits behind routine cleaning, and the guest waits at the desk.
- **Requests get lost.** A guest calls reception for towels; reception calls the supervisor, who looks for a free housekeeper. Each hand-off adds delay.
- **Nobody knows who is free.** The supervisor cannot see which staff are busy on which floor, so one floor is overloaded while another is idle.
- **Reception is blind.** The desk cannot see whether a room is dirty, being cleaned or ready without phoning housekeeping.
- **Guests get no feedback.** After asking, a guest cannot tell whether anyone accepted the request.

2 Who is affected

The first target is a mid-size business hotel of about 200 rooms over several floors, with housekeeping run in shifts and no costly property-management system. Five roles use the tool.

Role	Responsibility	Pain today	Can see
Housekeeping staff	Accept or decline jobs, do the work, update room status	Work arrives by phone or shouting, with no order	Only their assigned or eligible jobs
Guest	Raise a request and follow its status	Calls the desk and hears nothing back	Only their own request
Housekeeping manager	Allocate staff to floors, watch workload, handle escalations	Cannot see who is free or what is waiting	All jobs, staff load, decline reasons
Receptionist	Check guests in and out; tell arrivals when rooms are ready	Must phone housekeeping for room status	Room readiness; no staff administration
Administrator	Approve staff accounts and roles	Shared logins, no control over access	Accounts and roles

3 The solution

SmartKeep is a responsive web application. It ranks every housekeeping job by a visible priority score and sends a browser push alert to free staff on the right floor; the first to accept owns the job, and everyone sees its status live.

1. Reception checks a guest in (phone number, room, arrival time, VIP flag) or out. A checkout marks the room *Dirty* and creates a cleaning job.
2. A checked-in guest opens the guest page, enters their phone number, and the system finds their room. They pick **Maintenance**, **Cleaning** or **Extra items**. No app or login is needed.
3. The server scores the job and adds it to the manager's live queue in score order, with the score explained.
4. Every free housekeeper on that room's floor gets a push alert. If that floor is fully busy, the nearest floor with free staff is alerted.
5. The first to accept owns the job; others may decline with a reason only the manager sees.

6. The housekeeper taps *Start cleaning*, then *Done*. The room turns *Clean* on reception's board and the guest sees *Done*.

Two status flows run side by side: a request moves **Pending → Accepted → Done**, and a room moves **Dirty → Cleaning → Clean**. Only cleaning jobs change room status.

4 How jobs are prioritised

Each job gets a score from fixed rules, not AI, so the order is predictable and can always be explained. All weights are settings the hotel can tune.

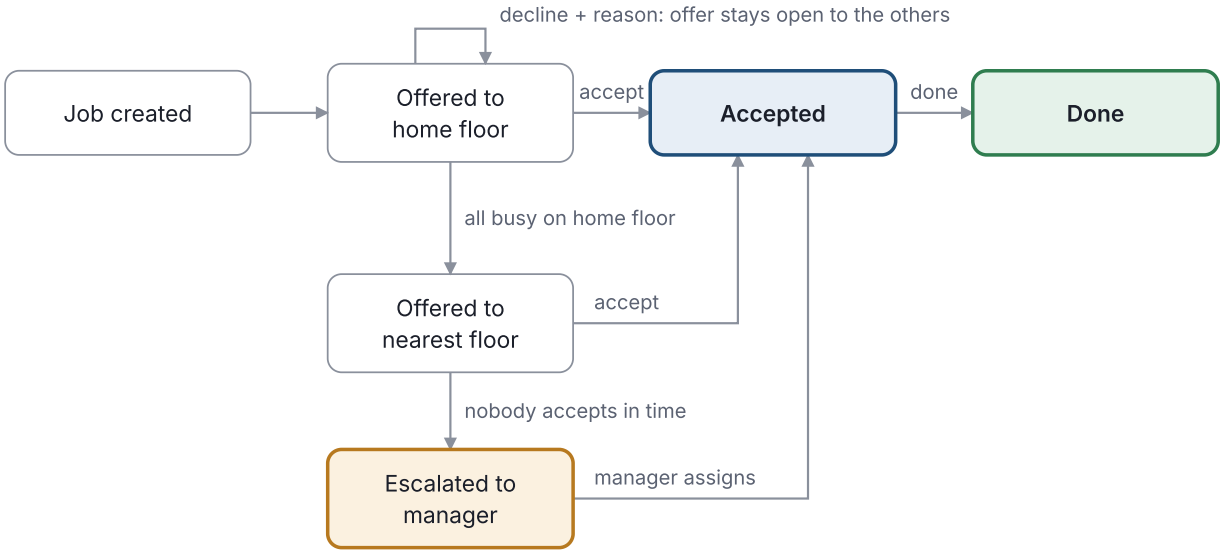
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Score = Service base + Arrival boost + VIP bonus + Waiting bonus
```

Factor	Points	Why
Maintenance (water, AC, breakage)	40	Guest cannot use the room properly
Cleaning (toilet, room)	30	Direct guest request
Extra items (towel, bedsheet)	20	Comfort request
Checkout cleaning	10	Room is empty; urgent only if someone is arriving
A guest arrives in this room within 2 hours	+50	Arrival-due rooms must beat everything
VIP guest	+15	Flag set by reception
Waiting time	+1 per 5 min, max +20	No request is forgotten

> **Worked example.** A checkout room with a guest due in an hour scores 10 + 50 = **60** and goes above a maintenance request at 40. A VIP arrival in the same room scores 75.

5 How jobs reach staff

Staff are allocated to floors per shift. A staff member is *free* when on shift and holding no accepted job; each person works one job at a time.



Request lifecycle. A decline never moves the job on; the manager is involved only when no free staff on the home or nearest floor accept.

- **Floor rule:** floor = room number ÷ 100 (room 214 → floor 2). A tie between the floor above and below alerts both.
- **Decline reasons:** Finishing another room · On break · Needs maintenance skills · Other (typed).
- **Push only:** a housekeeper cannot go on shift until browser notifications are allowed, so no alert is missed silently.

6 Core features

Three Must-Have features make up the first release; each has testable acceptance criteria in the product backlog (13 user stories, 4 epics).

ID	Feature	Key acceptance checks
F-01	**Guest request and live priority queue**	Known phone shows the room and 3 services; unknown phone says “please contact reception”; duplicate requests blocked; queue re-sorts within 5 s with score breakdowns.
F-02	**Floor alerts with nearest-floor fallback**	Only free staff on the floor get the push within 5 s; first accept wins and the other offers close; busy floor falls back to the nearest; decliners are not re-alerted.
F-03	**Room status Dirty → Cleaning → Clean**	Checkout marks the room Dirty and queues a job; reception’s board updates within 5 s; Done frees the housekeeper; extra-items jobs never change room status.

Next in line: manager workload view and floor allocation, reception check-in form with arrival time and VIP, role logins, timeout escalation, administrator approval.

7 Screens by role

Role	Screens
Guest	Phone entry → Service picker → My request status
Housekeeping staff	Enable notifications → My floor, on shift → Job offer (Accept / Decline + reason) → Active job (Start cleaning / Done)
Housekeeping manager	Live priority queue with score breakdown · Staff load per floor · Escalations · Floor allocation
Receptionist	Room board (all rooms by floor, coloured by status) · Check-in / checkout form
Administrator	Pending accounts → Approve / Reject

8 Technology

Layer	Choice	Why
Frontend	React (Vite), responsive, service worker	One web app for phones, tablets and desktops
Backend	Python FastAPI + SQLAlchemy	Built-in WebSockets; automatic API documentation
Database	PostgreSQL	Reliable, free, open source
Alerts	Web Push (VAPID) via pywebpush	Alerts reach the browser without an app install
Live screens	WebSockets	Queue, room board and guest status update without refresh
Packaging	Docker Compose	Whole system starts with one command

9 What makes it different

Today's option	How it works	Gap SmartKeep fills
Paper room sheet	Supervisor prints a list each morning	Goes stale within hours; no live requests or ranking
Phone calls / walkie-talkie	Reception → supervisor → housekeeper	Every hand-off adds delay; no record; no guest update
WhatsApp group	Requests posted to a staff chat	No order, no ownership; messages get buried
PMS housekeeping module	Room status inside hotel management software	Costly for mid-size hotels; often no guest self-service or floor routing (assumption)

- **Explainable priority:** every score is broken down on screen.
- **Arrival-aware:** rooms with guests arriving soon jump to the top automatically.
- **Floor-first routing:** the nearest free housekeeper gets the job; busy floors are covered.
- **No app, no login for guests:** a phone number is enough.

10 Success measures

Targets are the team's goals for the prototype, each tested live with seeded data; they are not measured industry figures.

Measure	Target
Push alert reaches staff after a guest submits	≤ 5 s
Queue order matches the score rules (10 seeded cases)	100%
Arrival-due room ranks above non-arrival jobs	Always
Fallback reaches the nearest floor when the home floor is busy	100%
Guest and reception see status changes	≤ 5 s
Clicks to accept a job, and to change room status	≤ 2 each

After a real pilot: average request-to-accept time, and the share of arrivals whose room is Clean on time.

11 Cost, rollout and scale

Estimated cost (assumptions)

Item	Estimate
Development tools	₹0 (open source)
Hosting, one hotel	₹1,000–2,500 / month
Extra devices	None
Setup and training	About 1 day
Shared service, per hotel	₹300–800 / month

Rollout

1. Pilot in one hotel for 2–4 weeks; load rooms and staff; tune weights.
2. Measure response times and staff feedback.
3. Fix what the pilot finds.

Scale

- Monthly web service shared by many hotels.
- Sync with booking and PMS systems.
- Tamil, Hindi and English staff screens.
- Hostels, hospitals and serviced apartments.

12 Assumptions and risks

Assumptions to validate

- A 200-room hotel runs about 15 housekeepers per shift, allocated by floor.
- The score weights and the 2-hour arrival window reflect real urgency.
- Staff have a phone, tablet or PC with internet on every floor.
- Each phone number maps to one active stay, and guests will use a web page instead of calling.

Risk	Mitigation
Browser push blocked or misconfigured	Notifications blocker before going on shift; push built and tested first
Phone-only lookup allows prank requests	Guest sees only room number and own status; manager can close a job
Score weights questioned	Breakdown shown on screen; weights are settings

Out of scope for the first release: PMS integration, payments, repair tracking, inspection step, SMS, multi-property and analytics.